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Research Areas

My research spans over areas of formal methods and software engineering. Specifically, I am interested in static and dynamic program analysis, verification and optimization of numerical programs, and automated systematic testing.

Education

- 11/2017 - 10/2023 **Dr.rer.nat, Technical University of Munich (TUM)**, Germany
Thesis: Improving Analysis and Optimization of Numerical Programs
Advisors: Helmut Seidl, Eva Darulova
Reviewers: Helmut Seidl, Eva Darulova, Zachary Tatlock
- 04/2015 - 07/2017 **Master of Science, University of Saarland**, Germany
Thesis: Estimation of Relative Error Bounds for Floating-Point Arithmetic Expressions
Advisor: Eva Darulova
Reviewers: Eva Darulova, Bernd Finkbeiner
GPA: 1.5
- 09/2008 - 07/2013 **Dipl.Engineer, St.Petersburg State Institute of Technology (Technical University)**, Russia
Major: Automated Information Processing and Control Systems
GPA: 4.91 of 5.0

Academic Positions

- 11/2023 - 07/2026 **Postdoctoral Researcher, TU Wien**, Vienna, Austria
Lab: Rigorous Software Engineering, led by Maria Christakis
- 11/2017 - 10/2023 **Scientific Employee, TUM**, Munich, Germany
Lab: Programming Languages, Compiler Construction and Specification Formalisms, led by Helmut Seidl
- 08/2018 - 10/2018 **Research Intern, Amazon Web Services**, New York City, USA
Lab: Automated Reasoning Group, mentor Kasper Luckow
- 08/2017 - 10/2017 **Scientific Employee, fortiss GmbH**, Munich, Germany
Department: Software Dependability, manager Holger Pfeifer

07/2016 - 07/2017 **Research Intern, Max-Planck Institute for Software Systems,**
from April 2017 Doctoral Student; Saarbrücken, Germany
Lab: Automated Verification & Approximation, led by Eva Darulova

Professional Experience

04/2013 - 04/2015 **Software Developer, T-Systems RUS,** St.Petersburg, Russia

Teaching and Mentoring Experience

Teaching at TU Wien

Software Engineering Seminar	summer 2024, 2025, winter 2024/25, 2025/26
Advanced Software Engineering (<i>guest lecture</i>)	winter 2025/26

Teaching at TUM

Lectures (teaching assistant)

Program Optimization	winter 2018/19, 2019/20, 2020/21, 2021/22, 2022/23
Virtual Machines	summer 2019, 2020, 2021, 2022, 2023
Functional Programming and Verification	winter 2017/18

Seminars

Program Synthesis (organizer)	summer 2018, 2019, 2020
The Tyranny of Types: Curse or Blessing?	summer 2022
Code Generation and Innovative Programming Models	winter 2019/20

Mentoring

SIGPLAN-M Longterm Mentoring Program	mentor for 1 PhD student, 2021 - <i>present</i>
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Co-advised theses

Interactive Program Synthesis for Object-Oriented Programs, TUM, *Master Thesis* of A.Semin, 2022
Regime Inference for Finite-Precision Kernels, TUM, *Bachelor Thesis* of R.Rabe, 2020

Academic Service

Co-Organizer	FLoC 2026 Mentoring Workshop
Publicity Chair	International Symposium on Software Testing and Analysis (ISSTA) 2024

Program Committee Member	Object-Oriented Programming, Systems, Languages & Applications (OOPSLA) 2026 Computer-Aided Verification (CAV) 2025 Programming Language Design and Implementation, Student Research Competition (PLDI SRC) 2024 Static Analysis Symposium (SAS) 2024, 2023, 2020 Automated Software Engineering, Tool Demo Track 2024 Formal Aspects of Computing (FAC) Journal 2021, 2026
Hiring Committee Member	TU Wien, Professorship for Programming Languages and Compilers 2024 TU Wien, Professorship for Program Analysis and Verification 2025
Artifact Evaluation Committee Member	Programming Language Design and Implementation (PLDI) 2023 International Symposium on Code Generation and Optimization (CGO) 2022 Computer-Aided Verification (CAV) 2021 Principles of Programming Languages (POPL) 2021
Student Volunteer	Federated Logic Conference (FLoC) 2022 European Joint Conferences on Theory and Practice of Software (ETAPS) 2022 Dagstuhl Seminar on Approximate Systems 2021 Principles of Programming Languages (POPL) 2020 Marktoberdorf Summer School 2019
Scientific Staff Representative	TUM, representing the lab of Programming Languages, Compiler Construction and Specification Formalisms, 2020 - 2023

Publications

Note: publications before 2023 use the old spelling of my name: Anastasiia Izycheva.

13. **Interrogation Testing of CHC Solvers**, FSE'26. (*Preprint*)
David Kaundstorfer, *Anastasia Isychev*, Valentin Wüstholtz, Maria Christakis
12. **Cost of Soundness in Mixed-Precision Tuning**, OOPSLA'25. (*DOI*)
Anastasia Isychev, Debasmitta Lohar
11. **Using Action-Policy Testing in RL to Reduce the Number of Bugs**, SoCS'25. (*DOI*)
Hasan Ferit Eniser, Songtuan Lin, Nicola Müller, *Anastasia Isychev*, Valentin Wüstholtz, Isabel Valera, Jörg Hoffmann, Maria Christakis

10. **Lazy Testing of Machine-Learning Models**, IJCAI'25. (DOI)
Anastasia Isychev, Valentin Wüstholtz, Maria Christakis
9. **Fuzzing Processing Pipelines for Zero-Knowledge Circuits**, CCS'25. (DOI)
Christoph Hochrainer, Anastasia Isychev, Valentin Wüstholtz, Maria Christakis
8. **Interrogation Testing of Program Analyzers for Soundness and Precision Issues**, ASE'24. (DOI)
David Kaindlstorfer, Anastasia Isychev, Valentin Wüstholtz, Maria Christakis
7. **Constraint-Based Test Oracles for Program Analyzers** ASE'24. (DOI)
Markus Fleischmann, David Kaindlstorfer, Anastasia Isychev, Valentin Wüstholtz, Maria Christakis
6. **Scaling up Roundoff Analysis of Functional Data Structure Programs**, SAS'23. (DOI)
Anastasia Isychev, Eva Darulova
5. **Regime Inference for Sound Floating-Point Optimizations**, EMSOFT'21. (DOI)
Robert Rabe, Anastasiia Izycheva, Eva Darulova
4. **Counterexample- and Simulation-Guided Floating-Point Loop Invariant Synthesis**, SAS'20. (DOI)
Anastasiia Izycheva, Eva Darulova, Helmut Seidl
3. **Synthesizing Efficient Low-Precision Kernels**, ATVA'19. (DOI)
Anastasiia Izycheva, Eva Darulova, Helmut Seidl
2. **Daisy - Framework for Analysis and Optimization of Numerical Programs** (Tool Paper), TACAS'18. (DOI)
Eva Darulova, Anastasiia Izycheva, Fariha Nasir, Fabian Ritter, Heiko Becker, Robert Bastian
1. **On Sound Relative Error Bounds for Floating-Point Arithmetic**, FMCAD'17. (*conference version, arXiv extended version*)
Anastasiia Izycheva, Eva Darulova

Invited Presentations

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| 2026 | Who Guards The Guards? Testing Program Analyzers , Helmutfest, Munich (<i>upcoming</i>) |
| 2026 | Cost of Soundness in Mixed-Precision Tuning , FPTalks, online |
| 2026 | Towards Reliable and Efficient Numerical Code through Sound Optimizations , Inria Lille, APR LIP6 & Inria Paris |
| 2021 | Synthesizing Efficient Low-Precision Kernels , FPTalks, online (recording available) |
| 2019 | Synthesizing Efficient Low-Precision Kernels , Marktoberdorf Summer School (short talk and poster) |
| 2018 | Polynomial Approximations in Numerical Kernels , Google Compiler and Programming Languages Summit 2018 (poster), Munich |

2018 **Daisy - a Framework for Sound Accuracy Analysis and Optimization of Numerical Programs**, Metalibm Workshop, Paris

Open-Source Projects

LaZ Framework for lazy testing of machine-learning models

DS2L Scalable sound rounding error analysis for numerical programs with data structures

Pine Invariant generator for floating-point loops

Daisy A framework for analysis and optimization of numerical programs

Scholarships and Grants

2025 SIGPLAN PAC travel grant for attending OOPSLA

2020 Travel grant for attending VMCAI Winter School

2020 SIGPLAN PAC travel grant for attending POPL

2019 Travel grant, Verification Mentoring Workshop (could not attend due to visa issues)

2017 Doctoral student scholarship, Max-Planck Institute for Software Systems

2015 - 2017 Doctoral student scholarship, Graduate School of Computer Science, University of Saarland

2008 - 2013 Scholarship for outstanding academic performance, St.Petersburg State Institute of Technology

Programming Skills

Advanced: Scala, Python, SQL

Intermediate: C, Java

Language Proficiency

Advanced: English, German

Beginner: Spanish

Native: Russian